

Advanced Data Management

Query Answering in GAV

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Outline of the course

1. Introduction to propositional logic
 2. Introduction to First-order Logic
 3. Relational Calculus
 4. Information Integration **S**ystems (IIS)
 5. Logical formalization of IISs
 6. Mapping between **Global Schema** e **Data Sources**
 7. Incomplete information databases
 8. **Query answering over IISs**
 9. **O**ntology-**B**ased **D**ata **M**anagement (OBDM)
 10. Description Logic Ontologies
 11. Query answering in ontologies
 12. Query answering in OBDM
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Query Answering in Information Integration Systems

Logical Formalization of Information Integration Systems – Syntax

An information integration system (IIS) is a triple $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ where:

- $\mathbf{G}$ is the **global schema**
 - A logical theory over a relational alphabet A_G .
- $\mathbf{S}$ is the **source schema**
 - A logical theory over a relational alphabet A_S .
- $\mathbf{M}$ is the **mapping between $\mathbf{S}$ and $\mathbf{G}$**
 - A logical theory over the relational alphabet $A_S \cup A_G$.

Logical Formalization of I.I.S.s – Semantics

Assume a information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is an information integration specification and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

The semantics of $\langle \mathbf{J}, \mathbf{D} \rangle$ is defined as follows:

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid \mathbf{B} \text{ is a global database s.t. } (\mathbf{B}, \mathbf{D}) \models \mathbf{M} \}$$

Assume a query q over A_G . The **certain answers** of q w.r.t. $\mathbf{J}$ and $\mathbf{D}$ are:

$$\text{cert}(q, \mathbf{J}, \mathbf{D}) = \{ \bar{a} \mid \bar{a} \in q(\mathbf{B}) \text{ for each } \mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D}) \}$$

Query Answering – Recognition Problem

- **DEF:** **QA-IIS** is the following decision problem:
 - **Input:** an I.I.S. $J = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$, a FOL query $q(\bar{x})$, a source database D , a tuple of constants $\bar{c}$
 - **Question:** Is $\bar{c} \in \text{cert}(q, J, D)$?
- **Thm:** The problem **QA-IIS** is **undecidable**.
- **Proof:** Reduction from Validity of FOL formulae.

Conjunctive Query (recap)

- Let $\varphi(x_1, \dots, x_n)$ be a FOL formula with free variables $x_1, \dots, x_n$
- $\varphi(x_1, \dots, x_n)$ is an Existential Conjunction if it is of the form

$$\exists y_1, \dots, \exists y_n P_1(\bar{z}_1) \wedge \dots \wedge P_k(\bar{z}_k)$$

- A FOL query $\{(x_1, \dots, x_n) | \varphi(x_1, \dots, x_n)\}$ is a **Conjunctive Query** if $\varphi(x_1, \dots, x_n)$ is an Existential Conjunction.

Union of Conjunctive Queries (recap)

- A formula $\varphi(x_1, \dots, x_n)$ is Union of Existential Conjunctions if it is of the form

$$\bigvee_i \psi_i(x_1, \dots, x_n)$$

where each ψ_i is an existential conjunction.

Observe: same free variables.

- A FOL query $\{(x_1, \dots, x_n) \mid \varphi(x_1, \dots, x_n)\}$ is a **Union of Conjunctive Queries** if $\varphi(x_1, \dots, x_n)$ is a Union of Existential Conjunctions.

Additional Assumptions

To restore decidability of query answering we assume:

- That **G** is an empty theory over the alphabet A_G , i.e., **G** is a database schema without integrity constraints.
- That **S** is an empty theory over the alphabet A_S , i.e., **S** is a database schema without integrity constraints.
- Well behaved mappings: LAV or GAV mappings defined via queries that we can actually handle: **Conjunctive Queries**

Some small relaxations may be possible (see later)

From now on, if not otherwise stated, we always consider the assumptions made above to be true moreover, if not otherwise stated, we will assume that the **user queries are **conjunctive queries****

(Easy) Query Answering – Recognition Problem

- A class of mapping is a family of finite logical theories (defining mappings)
- **DEF: UCQ-QA-IIS($\mathcal{L}$)** where $\mathcal{L}$ is a class of mappings is the following problem
 - **Input:** An I.I.S. $J = \langle G, M, S \rangle$ where G and S are empty theories and M is a mapping in $\mathcal{L}$, a source database D , a UCQ q , and a tuple of constants $\bar{c}$
 - **Question:** Is $\bar{c} \in \text{cert}(q, J, D)$?
- As we will see, **UCQ-QA-IIA($\mathcal{L}$)** is **decidable** in several interesting cases:
 - Conjunctive GAV mapping (CQ-GAV)
 - Conjunctive LAV mapping (CQ-LAV)
 - Conjunctive GLAV mapping (CQ-LAV)

Techniques for Query Answering

Techniques to solve **UCQ-QA-IIS**($\mathcal{L}$) can be divided into two families:

- **Materialization-based Approaches**
 - A special *instance* of the **global schema** is computed independently from the query
 - We compute the certain answers by evaluating the input query over such instance
 - Useful to materialize the result of the integration (**data exchange**)!
- **Virtualization-based Approaches**
 - Queries over the **global schema** are translated into suitable queries over the **source schema**
 - We compute the certain answers by evaluating the rewritten query directly over the **source database**
 - We cannot use this to materialize the result of the integration but data is always fresh!

Query Answering In Information Integration

	Materialized	Virtual
GAV	1	2
LAV	3	4

GAV Mapping (recap)

Global As View (recap)

GAV mappings describe the **global schema** in terms of the **source schema**

A GAV **mapping assertion** is a pair $\langle q_s, g \rangle$ where:

- $q_s = \{\bar{x} \mid \varphi_s(\bar{x})\}$ is a FOL **query of arity n** over A_s
- g is a **single predicate** name of arity **n** in A_g

Logical formalization of GAV mapping assertions

$$\forall \bar{x} . \varphi_s(\bar{x}) \rightarrow g(\bar{x})$$

A GAV mapping is a **finite** set of GAV mapping assertions.

Conjunctive GAV (CQ-GAV)

GAV mappings describe the **global schema** in terms of the **source schema**

A **Conjunctive GAV** mapping assertion is a pair $\langle q_s, g \rangle$ where:

- $q_s = \{\bar{x} \mid \varphi_s(\bar{x})\}$ is a **conjunctive query** of arity n over A_s
- g is a single predicate name of arity n in A_G

Logical formalization of **Conjunctive GAV** mapping assertions

$$\forall \bar{x} . \varphi_s(\bar{x}) \rightarrow g(\bar{x})$$

A **Conjunctive GAV** mapping is a **finite** set of **Conjunctive GAV** mapping assertions.

Query Answering in GAV

Query Answering In Information Integration

	Materialized	Virtual
GAV	1	2
LAV	3	4

Materialization in GAV

- Assume an I.I.S $J = \langle G, M, S \rangle$ and a **source database D** for **S**
- We want to produce a database $M(D)$ to represents the result of the integration
- Desired properties of the materializations.
 - We want to use the materialization $M(D)$ to compute **certain answers** to queries
 - We want the materialization to be as **compact** as possible
- **Idea**: to compute a materialization $M(D)$ of GAV mappings, we can simply materialize the views, i.e., the definitions of the atoms via the following steps:
 1. “Answer” over **D** the **queries** in the left-hand side of each **mapping assertion**,
 2. “Load” these answers into the **elements** of the **global schemas**

Retrieved Global Database

Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is a GAV information integration specification and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

We call **retrieved global database** for $\mathbf{J}$ w.r.t. $\mathbf{D}$, denoted by $\mathbf{M}(\mathbf{D})$, the **global database** obtained by "applying" the queries in the **mapping**, and "transferring" to the elements of $\mathbf{G}$ the corresponding tuples retrieved from $\mathbf{D}$

Formally:

$$\mathbf{M}(\mathbf{D}) = \{ \mathbf{g}(\bar{c}) \mid \langle \mathbf{q}_s, \mathbf{g} \rangle \in \mathbf{M} \text{ and } \bar{c} \in \mathbf{q}_s(\mathbf{D}) \}$$

Note that, since mappings are of type GAV, the tuples to be "transferred" to the **global schema** $\mathbf{G}$ are always definite, i.e., they do not contain existentially quantified elements.

Retrieved Global Database: Observations

Retrieved global database for $\mathbf{J}$ w.r.t. $\mathbf{D}$:

$$\mathbf{M}(\mathbf{D}) = \{g(\bar{c}) \mid \langle q_s, g \rangle \in \mathbf{M} \text{ and } \bar{c} \in q_s(\mathbf{D})\}$$

In general, given a source database $\mathbf{D}$, we saw that there are several **global databases** for $\mathbf{G}$ that satisfy $\mathbf{M}$ w.r.t. $\mathbf{C}$. However, it is easy to prove the following properties:

Property 1: $\mathbf{M}(\mathbf{D}) \in \text{sem}(\mathbf{J}, \mathbf{D})$

Property 2: $\mathbf{M}(\mathbf{D}) \subseteq \mathbf{B}$ for each $\mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D})$

That is, the **retrieved global database** is the *smallest* (w.r.t. set containment) **global database** in $\text{sem}(\mathbf{J}, \mathbf{D})$.

Retrieved Global Database - Example

- **Global Schema G** containing:

- Movie(Title, Year, Director)
- European(Director)
- Review(Title, Critique)

- **Source Schema S** containing:

- **r1(Title, Year)** : Movies since 1960 with European directors
- **r2(Title, Critique)** : Reviews of movies since 1990
- **r3(Title, Director, Country)** : Movies' directors with their country

GAV Mapping $M=\{m_1, m_2, m_3\}$:

- $m_1 = \forall t, y, d, c. r_1(t, y) \wedge r_3(t, d, c) \rightarrow \text{Movie}(t, y, d)$
- $m_2 = \forall t, d. r_3(t, d, \text{'EU'}) \rightarrow \text{European}(d)$
- $m_3 = \forall t, r. r_2(t, r) \rightarrow \text{Review}(t, r)$

Retrieved Global Database - Example

D

r₁

Title	Year
T1	1970
T2	1990
T3	1979

r₂

Title	Critique
T2	7/10

r₃

Title	Director	Country
T1	D1	EU
T2	D2	EU

M(D)

Movie

Title	Year	Director
T1	1970	D1
T2	1990	D2

$m_1 = \forall t, y, d, c. r_1(t, y) \wedge r_3(t, d, c) \rightarrow \text{Movie}(t, y, d)$

Retrieved Global Database - Example

D

r₁

Title	Year
T1	1970
T2	1990
T3	1979

r₂

Title	Critique
T2	7/10

r₃

Title	Director	Country
T1	D1	EU
T2	D2	EU

M(D)

European

Director

D1

D2

$m_2 = \forall t, d. r_3(t, d, 'EU') \rightarrow \text{European}(d)$

Retrieved Global Database - Example

D

r₁

Title	Year
T1	1970
T2	1990
T3	1979

r₂

Title	Critique
T2	7/10

r₃

Title	Director	Country
T1	D1	EU
T2	D2	EU

M(D)

Review

Title	Critique
T2	7/10

$$m_3 = \forall t, r. r_2(t, r) \rightarrow \text{Review}(t, r)$$

Retrieved Global Database - Example

D

r₁

Title	Year
T1	1970
T2	1990
T3	1979

r₂

Title	Critique
T2	7/10

r₃

Title	Director	Country
T1	D1	EU
T2	D2	EU

M(D)

Movie

Title	Year	Director
T1	1970	D1
T2	1990	D2

European

Director
D1
D2

Review

Title	Critique
T2	7/10

Retrieved Global Database for Query Answering with CQ-GAV mapping

Thm: Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is a CQ-GAV information integration specification and $\mathbf{D}$ is a **source database**. For any CQ q over $\mathbf{G}$, we have $\text{cert}(q, \mathbf{J}, \mathbf{D}) = q(\mathbf{M}(\mathbf{D}))$

Exercise: Prove the Theorem

Hints: First prove that $\text{cert}(q, \mathbf{J}, \mathbf{D}) \subseteq q(\mathbf{M}(\mathbf{D}))$ by contraposition, i.e. prove that $\bar{c} \notin q(\mathbf{M}(\mathbf{D}))$ implies $\bar{c} \notin \text{cert}(q, \mathbf{J}, \mathbf{D})$ using **Property 1**

Then, prove that $q(\mathbf{M}(\mathbf{D})) \subseteq \text{cert}(q, \mathbf{J}, \mathbf{D})$ using **Property 2** + the fact that CQs are *monotone*, i.e. $q(\mathbf{B}) \subseteq q(\mathbf{B}')$ holds whenever q is a CQ and $\mathbf{B} \subseteq \mathbf{B}'$

Retrieved Global Database for Query Answering with CQ-GAV mapping

Thm: Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is a CQ-GAV information integration specification and $\mathbf{D}$ is a **source database**. For any CQ q over $\mathbf{G}$, we have $\text{cert}(q, \mathbf{J}, \mathbf{D}) = q(\mathbf{M}(\mathbf{D}))$

Proof: Suppose $\bar{c} \notin q(\mathbf{M}(\mathbf{D}))$. Since $\mathbf{M}(\mathbf{D}) \in \text{sem}(\mathbf{J}, \mathbf{D})$ (cf. **Property 1**), we have $\bar{c} \notin \text{cert}(q, \mathbf{J}, \mathbf{D})$ by definition of certain answers.

Suppose now $\bar{c} \in q(\mathbf{M}(\mathbf{D}))$. Since $\mathbf{M}(\mathbf{D}) \subseteq \mathbf{B}$ for each $\mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D})$ (cf. **Property 2**), and since CQs are *monotone*, i.e. $q(\mathbf{B}) \subseteq q(\mathbf{B}')$ holds for any $\mathbf{B} \subseteq \mathbf{B}'$, we derive $\bar{c} \in q(\mathbf{B})$ for any $\mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D})$. Hence, we have that $\bar{c} \in \text{cert}(q, \mathbf{J}, \mathbf{D})$.

Can we extend these results?

Retrieved Global Database for Query Answering with FOL-GAV mapping

Actually the theorem holds in case of general GAV mapping (FOL-GAV)!

Thm: Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is a **FOL-GAV** information integration specification and $\mathbf{D}$ is a **source database**. For any CQ $\mathbf{q}$ over $\mathbf{G}$, we have $\text{cert}(\mathbf{q}, \mathbf{J}, \mathbf{D}) = \mathbf{q}(\mathbf{M}(\mathbf{D}))$.

Proof: the same technique can be used also in case of general GAV mapping (FOL-GAV). The claim then follows from the fact that the retrieved global database has the same exact properties as in the case of CQ-GAV.

Query Answering In Information Integration

	Materialized	Virtual
GAV	1	2
LAV	3	4

Virtualization in GAV

- We want to produce a query over the **sources** that captures the results of a user query over the **global schema**.
- Desired properties of the source query.
 - We want to use it to compute **certain answers**.
 - It should fall into a specific query language.
 - It should have the right size.
- **Virtualization in GAV is very simple**: we replace atoms in the user query with their definition in the **mapping**.

Perfect Rewritings

- Assume an Information Integration System $J = \langle G, M, S \rangle$.
 - No further assumptions on the three components for now
- Assume a class of queries L (a language of queries), e.g., CQs, UCQs, FOL, ...
- **Definition:** Given a query q over G , a **perfect L -rewriting of q w.r.t. J** is a query $rew_L^{(q,J)}$ in the language L such that the following condition holds for every database D for S

$$\bar{a} \in cert(q, J, D) \text{ if and only if } \bar{a} \in rew_L^{(q,J)}(D)$$

Query Unfolding in CQ-GAV

The **unfolding** of a query q over G w.r.t. M (denoted by $unf_{q,M}$) is the query over S obtained from q by substituting every atom $R(\bar{z})$ in q such that $\langle \psi_S, R \rangle \in M$ with the corresponding definition $\psi_S(\bar{z})$.

The **unfolding** of a UCQ is the union of the unfolding of all its disjuncts.

- **Thm:** Given an information integration specification $J = \langle G, M, S \rangle$, where M is a CQ-GAV mapping, and given a UCQ q over G , we have that $unf_{q,M}$ is a **perfect UCQ-rewriting** of q w.r.t. J .
- **Corollary:** For any UCQ q over G , and **source database** D for S , we have $cert(q, J, D) = unf_{q,M}(D)$

Query Unfolding – Example (1\3)

- **Global Schema G** containing:

- Paper(ID, Title)
- Author(PaperID, authorID, position)
- Researcher(ID, Affiliation)

- **Source Schema S** containing:

- **T1(article, title, SSD)** : papers with their title and scientific sectors (e.g., chemistry, mathematic, etc.).
- **T2(article, author, university, pos)** : authors of scientific articles with their affiliation and position as author of the article.
- **T3(researcher, univ, year)** : affiliation and affiliation date of the researchers.

- **GAV Mapping M={m₁,m₂,m₃, m₄}**:

- **m₁** = $\forall a, t, s. T1(a, t, s) \rightarrow \text{Paper}(a, t)$
- **m₂** = $\forall ar, at, u, p. T2(ar, at, u, p) \rightarrow \text{Author}(ar, at, p)$
- **m₃** = $\forall ar, at, u, p. T2(ar, at, u, p) \rightarrow \text{Researcher}(at, u)$
- **m₄** = $\forall r, u, y. T3(r, u, y) \rightarrow \text{Researcher}(t, u)$

Query Unfolding – Example (2\3)

- GAV Mapping $M=\{m_1, m_2, m_3, m_4\}$:

- $m_1 = \forall a, t, s. T1(a, t, s) \rightarrow \text{Paper}(a, t)$
- $m_2 = \forall ar, at, u, p. T2(ar, at, u, p) \rightarrow \text{Author}(ar, at, p)$
- $m_3 = \forall ar, at, u, p. T2(ar, at, u, p) \rightarrow \text{Researcher}(at, u)$
- $m_4 = \forall r, u, y. T3(r, u, y) \rightarrow \text{Researcher}(t, u)$

User query over **G** asking for "*the papers together with their authors and affiliation*":

$$q = \{(p, r, a) \mid \exists n. \text{Researcher}(r, a) \wedge \text{Author}(p, r, n)\}$$

$unf_{q, M}$ is the union of the following two CQs over **S**:

- $\{(p, r, a) \mid \exists x, y, z, k. T2(x, r, a, y) \wedge T2(p, r, z, k)\}$
- $\{(p, r, a) \mid \exists x, y, z. T3(r, a, x) \wedge T2(p, r, y, z)\}$

Query Unfolding – Example (3\3)

■ GAV Mapping $M=\{m_1, m_2, m_3, m_4\}$:

- $m_1 = \forall a, t, s. T1(a, t, s) \rightarrow \text{Paper}(a, t)$
- $m_2 = \forall ar, at, u, p. T2(ar, at, u, p) \rightarrow \text{Author}(ar, at, p)$
- $m_3 = \forall ar, at, u, p. T2(ar, at, u, p) \rightarrow \text{Researcher}(at, u)$
- $m_4 = \forall r, u, y. T3(r, u, y) \rightarrow \text{Researcher}(t, u)$

User query over **G** asking for "the papers together with their authors and affiliation":

$$q = \{(p, r, a) \mid \exists n. \text{Researcher}(r, a) \wedge \text{Author}(p, r, n)\}$$

$unf_{q, M}$ is the union of the following two CQs over **S**:

- $\{(p, r, a) \mid \exists x, y, z, k. T2(x, r, a, y) \wedge T2(p, r, z, k)\}$
- $\{(p, r, a) \mid \exists x, y, z. T3(r, a, x) \wedge T2(p, r, y, z)\}$

Unfolding

FOL-GAV Mappings

Unfolding can be used also with more expressive (FOL-)GAV mappings!

– In the case of general GAV mappings the algorithm works as before

- **Thm:** Given an information integration specification $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$, where $\mathbf{M}$ is a FOL-GAV mapping, and given a UCQ q over $\mathbf{G}$, we have that $unf_{q, \mathbf{M}}$ is a **perfect FOL-rewriting** of q w.r.t. $\mathbf{J}$.

Observe: in this case, $unf_{q, \mathbf{M}}$ is a FOL query instead of a UCQ

Query Answering – Recognition Problem

Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is a CQ-GAV information integration specification (i.e., $\mathbf{M}$ is a CQ-GAV mapping) and $\mathbf{D}$ is a **source database**.

We know that $\mathit{unf}_{q, \mathbf{M}}$ is a UCQ that is the **perfect UCQ-rewriting** of q w.r.t. $\mathbf{J}$ and so we have that a tuple of constants $\bar{a} \in \mathit{cert}(q, \mathbf{J}, \mathbf{D})$ if and only if $\bar{a} \in \mathit{unf}_{q, \mathbf{M}}(\mathbf{D})$. From these observations, and from the fact that query answering of a UCQ (FOL-query) over a database is in LogSpace, we have:

Thm: UCQ-QA-IIS(CQ-GAV) is polynomial in **data complexity**, actually LogSpace.

This result holds also in case of the FOL-GAV class of mapping, i.e., UCQ-QA-IIS(FOL-GAV) is polynomial (in LogSpace) in **data complexity**.

Materialization vs Virtualization in GAV

- The size of the Retrieved Global Database depends exponentially on the size of the arity of predicates in the global schema!
 - It is very easy to compute the unfolding.
 - Should we always use virtualization?
 - Do we want to save a snapshot of the data?
 - Is accessing sources as easy as accessing a retrieved global database?
 - The answer to the above question may depend on the scenario of use.
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Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.
